

Exercise-AWS Overview

Q1)List out the types of instance base on the pricing model and write a brief about your understanding about it.

AWS EC2 instances can be classified by pricing model as follows:

1. On-Demand Instances

- Description: Pay for compute capacity by the second or hour with no long-term commitment.
- This model is best when workloads are unpredictable or temporary. Since there is no upfront cost, it is ideal for testing, learning, or applications where usage cannot be forecasted. However, it is the most expensive option if used continuously.

2. Reserved Instances (RI)

- Description: Commit to AWS for 1 year or 3 years in exchange for lower pricing.
- Types:
 - 1)Standard RI (highest discount)
 - 2)Convertible RI (can change instance family)
- Reserved Instances are suitable for stable and long-running production workloads. By committing in advance, we reduce costs substantially. The downside is reduced flexibility, as we are locked into a commitment period.

3. Savings Plans

- Description: Commit to a consistent amount of usage (\$/hour) rather than a specific instance.
- Types:
 - 1)Compute Savings Plan
 - 2)EC2 Instance Savings Plan

- Savings Plans are more **flexible than Reserved Instances** because they allow changes in instance family, size, or region. This model works well for organizations with **consistent overall usage** but evolving architectures.

4. Spot Instances

- Description: Use unused AWS capacity at up to 90% discount.
- Spot Instances are extremely **cost-effective** but can be interrupted at any time. Therefore, they are suitable only for **fault-tolerant workloads** like batch processing, data analysis, and CI/CD pipelines.

5. Dedicated Hosts

- Description: Physical EC2 server dedicated to one customer.
- This model is mainly used for compliance, regulatory requirements, or custom licensing. It offers maximum control over hardware but comes at a high cost, making it unsuitable for general workloads.

6. Dedicated Instances

- Description: Instances run on hardware dedicated to a single customer.
- Dedicated Instances offer **hardware-level isolation** without giving full control of the physical server. They are a middle ground between shared tenancy and Dedicated Hosts, mainly used for security and compliance needs.

Q2) Host a static website in S3.

Steps to host a static website in S3:

Step1) Go to S3->Create bucket->Enter Bucket Name

Amazon S3 > Buckets > Create bucket

Create bucket

Buckets are containers for data stored in S3.

General configuration

AWS Region

US East (Ohio) us-east-2

Bucket type

☒ General purpose

☐ Directory

General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name

my-static-site

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ ACLs disabled (recommended)

☐ ACLs enabled

ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Step2)Upload Files->I uploaded index.html & logo.png

Amazon S3

Buckets

my-static-site-1343

Upload

Upload

Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (2 total, 11.8 KB)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

< 1 >

☐	Name	Folder	Type	Size
☐	index.html	-	text/html	872.0 B
☐	logo.png	-	image/png	10.9 KB

Destination

Info

Destination

[s3://my-static-site-1343](#)

Destination details

Bucket settings that impact new objects stored in the specified destination.

Permissions

Grant public access and access to other AWS accounts.

Properties

Specify storage class, encryption settings, tags, and more.

Cancel

Upload

Step3)Enable Static Website Hosting

aws

Search

[Alt+S]

United States (Ohio)

HARSH RAJ (2216-8301-3469)

HARSH RAJ

Amazon S3

Buckets

my-static-site-1343

Edit static website hosting

Edit static website hosting

Info

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☐ Disable

☒ Enable

Hosting type

☒ Host a static website

Use the bucket endpoint as the web address. [Learn more](#)

☐ Redirect requests for an object

Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

Specify the home or default page of the website.

index.html

Error document - optional

This is returned when an error occurs.

error.html

Step4) Make the Bucket Public

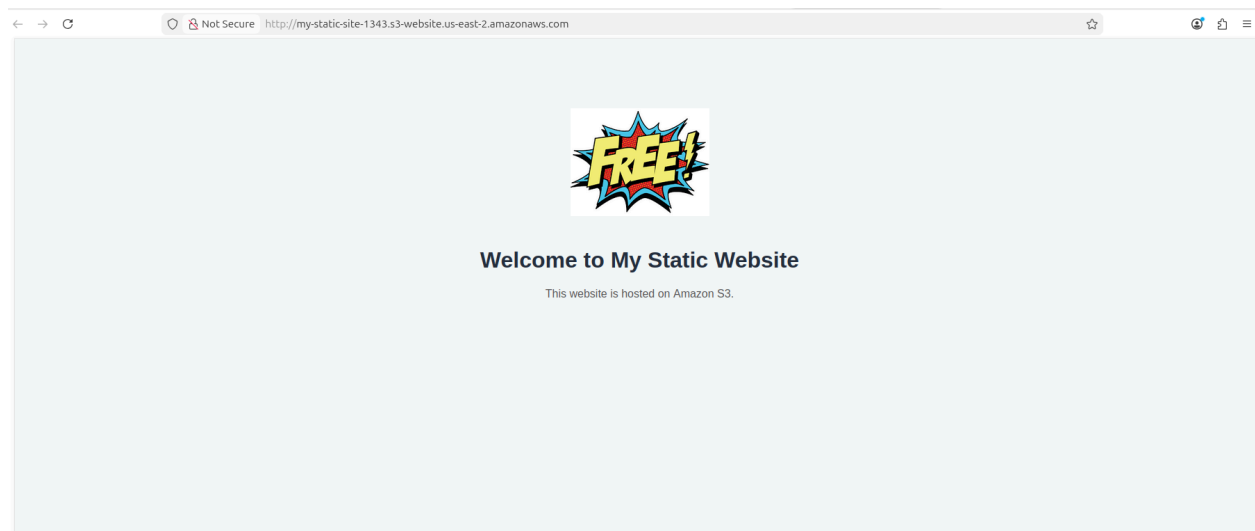
Go to Permissions Tab->Click Bucket Policy->Paste the Policy as shown in image

The screenshot shows the AWS IAM console interface for editing a bucket policy. The breadcrumb navigation at the top indicates the path: Amazon S3 > Buckets > my-static-site-1343 > Edit bucket policy. The main heading is 'Edit bucket policy' with an 'info' link. Below this, the 'Bucket policy' section explains that the policy is written in JSON and provides access to objects in the bucket. A 'Policy examples' link is available. The 'Bucket ARN' is shown as 'arn:aws:s3::my-static-site-1343'. The 'Policy' section contains a JSON policy document with a single statement that allows public read access. The JSON is as follows:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "allow",
7       "Principal": "**",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3::my-static-site-1343/**"
10    }
11  ]
12 }
13
```

On the right side, there is an 'Edit statement' section with a 'Select a statement' heading. It prompts the user to 'Select an existing statement in the policy or add a new statement.' and includes a '+ Add new statement' button.

Step5) Website is now live



Q3) Launch an Ubuntu EC2 instance on AWS, with 10GB root volume, and SSH from your local machine using the private key.

Steps to Launch Ubuntu EC2 instance:

Step1) Open EC2 console - Launch instance - Select Ubuntu EC2 instance- Select t2.micro

[EC2](#) > [Instances](#) > Launch an instance

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Quick Start


Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



Debian


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0f5fcd140e4ab7 (64-bit (x86)) / ami-07f75595710e1c42b (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble Image

Architecture	AMI ID	Publish Date	Username
64-bit (x86) ▼	ami-0f5fcd140e4ab7	2025-10-22	ubuntu

[Verified provider](#)

- Step2) Create a new key pair and download the private key (.pem file)
- Step3) Allow SSH-Set the root volume to 10 gb

EC2 > Instances > Launch an instance

▼ Storage (volumes) Info

Simple

EBS Volumes

Hide details

▼ Volume 1 (AMI Root) (Custom)

Storage type Info

EBS

Device name - required Info

/dev/sda1

Snapshot Info

snap-05fe089b7b96696dd

Size (GiB) Info

10

Volume type Info

gp3

Free tier eligible ▼

IOPS Info

3000

Delete on termination Info

Yes ▼

Encrypted Info

Encrypted ▼

KMS key Info

Select ▼

Throughput Info

125

Volume initialization rate - new, optional Info

Enter a value

Min: 100 MiB/s, Max: 300 MiB/s. Additional charges apply

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

×

Add new volume

The selected AMI contains instance store volumes, however the Instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information

🔄

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

Show details

Step 4) Connect to the EC2 Instance

```
harsh-raj@TTNPL-harsh:~ $ cd Downloads
harsh-raj@TTNPL-harsh:Downloads $ chmod 400 exercise.pem
harsh-raj@TTNPL-harsh:Downloads $ ssh -i exercise.pem ubuntu@3.129.39.180
The authenticity of host '3.129.39.180 (3.129.39.180)' can't be established.
ED25519 key fingerprint is SHA256:05HkJarmom0VRCFJvDWuhf2aNXAHTfoxoiupgJstk+Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.129.39.180' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jan 20 11:28:40 UTC 2026

System load:  0.03               Processes:            114
Usage of /:   20.1% of 8.65GB    Users logged in:     0
Memory usage: 20%               IPv4 address for enX0: 172.31.1.235
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```


Q4) Install nginx package in the above server and access this page from your local browser using a domain name instead of IP address of the server.

Steps to install nginx in the above server and access it locally:

Step1) Login to the EC2 server (via SSH)

`ssh -i mykey.pem ubuntu@<EC2_PUBLIC_IP>` , replace the <EC2_PUBLIC_IP> with your instance public key.

Step2) Install , Start and Enable Nginx

```
ubuntu@ip-172-31-1-235:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 86 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 0s (20.6 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 71735 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-1-235:~$ sudo systemctl start nginx
ubuntu@ip-172-31-1-235:~$ sudo systemctl enable nginx
```

Step3) Edit hosts file on your local machine

Sudo vi /etc/hosts

```
127.0.0.1 localhost
127.0.1.1 TTNPL-harsh
18.225.55.97 myserver.local
#
# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

Step4) Access website using Domain name (in my case “myserver.local”)

